
Lecture 1

COMP2211: Distributed Systems

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About me

- **Name:** Thiru Vasantam
- Member of NESTiD group
- **Research Interests:** Design of scheduling algorithms and performance evaluation of computer networks focusing on both classical and quantum networks;
- I work on quantum internet, cloud computing and data center models, and sensor networks etc.
- I use applied probability, queueing theory, optimization, and network analysis theory.

Lecture 1

- Course organisation
- Topic 1: Introduction
 - Definition
 - Properties
 - Design challenges
- Topic 2: Middleware

Assessment

- **Office hours:** Mondays 2-3pm in my office MCS2008
- Written Examination only
 - Scheduled in May or June, class-room exam.
 - A revision lecture will be held in the Easter Term.



Sessions

- **Lectures:** Mondays 11am-1pm in CLC2023
- **Practicals:** Weeks 31-35, we have 5 practicals.

Main Reading

- **Main Reference:** Distributed Systems, 4th Edition, Maarten van Steen and Andrew Tannenbaum.

I will provide a link on Ultra to download a soft copy from authors website.

- **Distributed systems : concepts and design / George Coulouris ... [et al.],** Addison-Wesley (Pearson), c2012.

Changes This Year Based on the Last Year MEQs

- Last year, I introduced a lot of new content, and students advised me to improve the slides' structure. I am making these changes this year.
- Provide formative exercises
- Minor changes to content/resources

Topic 1

Topic 1: Introduction

- Definition
- Properties
- Design challenges

Recap of Networks

- Study standards and methods to connect machines for communication

Standards

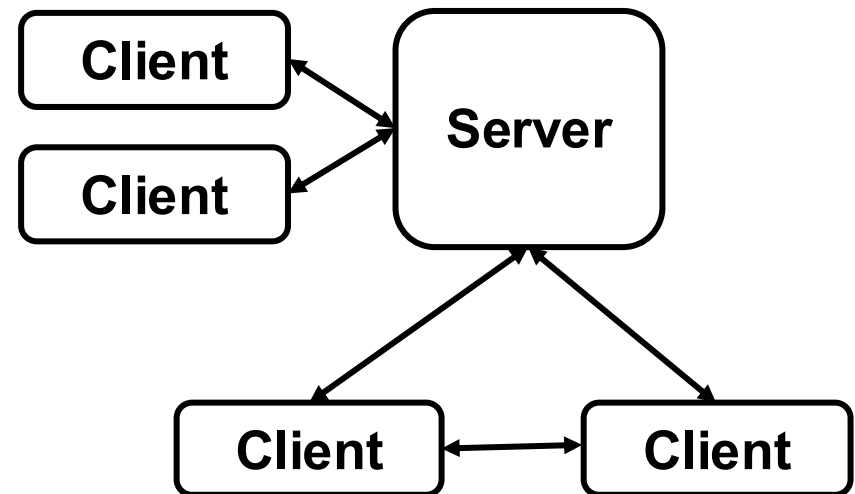
- OSI Model (Network layers), IP Addressing (IPv4, IPv6),
Protocols (communication methods)

Machine Addressing

- Host ID, Port number,
- Unicasting, multicasting

Machine roles

- Client
- Server



Centralised System

- Has a single point of control for its operation
 - **Centralised services:** single server provides service to all users
 - **Centralised data:** single component stores all the data, ex: telephone book
 - **Centralised algorithm:** single component makes scheduling decisions, ex: routing based on complete information

Drawbacks:

- Single point of failure
- Difficult to scale up the system
- Centralised nature of the system: all customers must stay in a small region

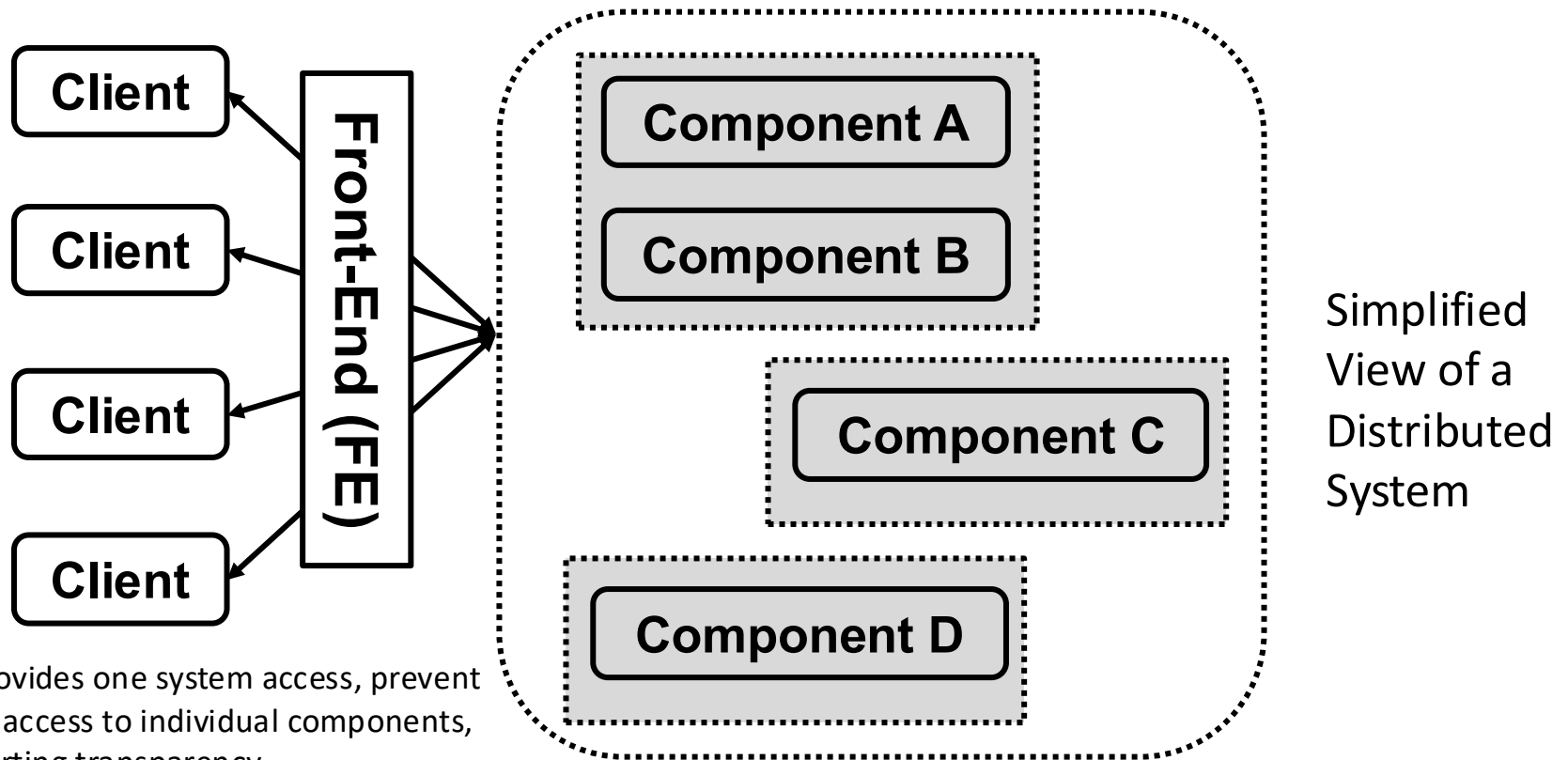
Definition: Distributed System

- Initial computer systems were small-scale and centralised, but modern systems are decentralised/distributed.
- **Distributed system:** It is a collection of independent and dynamic components, which work together and appear to its users as a single coherent system. Components can be
 - Hardware (Servers, Sensors, Debit/Credit card readers, etc.)
 - Software components
 - Web Services
- It helps us to share resources of components through communication

Can you think of any examples of distributed systems?

Distributed Systems: Block Diagram

- In Distributed Systems, we study technologies that connect networked, geographically dispersed, dynamic components to form a powerful, high-performance, and reliable system.



FE: provides one system access, prevent direct access to individual components, supporting transparency

Difference Between Networks and Distributed Systems

Networks:

- Focus: method and implementation of communication
- Data transmission: addressing, routing, scheduling, quality control, and error recovery

Distributed systems (DS): Support application development

- Components: heterogeneous (OS, H/W, S/W), physically loosely coupled, but are required to collaborate
- System architecture: transparency, performance, reliability
- Network technology: a means to connect and organise DS components

Why Distributed Systems?

- Many real-world systems are distributed
 - World wide web, Google
 - Cloud computing
 - Amazon.com
 - Modern networked computers, ex: Netflix, Online Gaming
- Useful to understand how such real-world systems work
- This course is about designing distributed systems; will learn how to design future systems

Example: Hotel Booking

Booking.com: Hotels in Dur x

www.booking.com/searchresults.html?src=index&error_url=http%3A%2F%2Fwww.booking.com%2Findex.en-gb.html%3Fsid%3De04a36413eac9bct

Distance: 2 miles

Check-in date
Mon 23 January 2012

Check-out date
Wed 25 January 2012

☐ I don't have specific dates yet
☐ I am travelling on business

Room 1: 2 adults

[Change selection](#)

Search

Narrow results:

▼ Price (per night)

- ☐ £ 0 - £ 39
- ☐ £ 40 - £ 49
- ☐ £ 50 - £ 69
- ☐ £ 70 - £ 119
- ☐ £ 120 +

▼ Star Rating

- ☐ 2 stars 1 hotel
- ☐ 3 stars 7 hotels
- ☐ 4 stars 5 hotels
- ☐ 5 stars 1 hotel
- ☐ Unrated 1 hotel

▼ Hotel Type

- ☐ Hotel 9 hotels
- ☐ Guest accommodation 6 hotels

▼ Facility

- ☐ Wi-Fi/Wireless LAN 13 hotels
- ☐ Parking 15 hotels

Radisson Blu Hotel, Durham ★★★★★

Just 1 mile from Durham Rail Station and city centre, the Radisson BLU Durham has fantastic views of Durham Cathedral. It offers stylish rooms, free Wi-Fi, a restaurant, bar and modern health club.

Double Room £ 168

Sort by: Recommended ▼ Stars Distance Review Score Price

Radisson Blu Hotel, Durham ★★★★★
Durham • [Show map](#)

Just 1 mile from Durham Rail Station and city centre, the Radisson BLU Durham has fantastic views of Durham Cathedral. It offers stylish rooms, free Wi-Fi, a restaurant, bar and modern health club. [More](#)

Latest booking: January 15

Fabulous, 8.6
Score from 536 reviews

Book now

Price for 2 nights

Standard Double Room	Save 35%	Available	£ 258 - £ 168
FREE cancellation			
Standard Twin Room	FREE cancellation	Available	£ 218 - £ 168
Business Class Room	FREE cancellation		

Example: Cloud Computing

- Scalable Computation Farm



- Animoto.com (online video maker):
 - Started with 50 servers on Amazon EC₂
 - Growth of 25,000 users/hour
 - Needed to scale to 3,500 servers in 2 days
- Zoom is an example; it had to scale-up its systems massively during pandemic
- Modern computing jobs: graphs represent tasks to be completed for a job, require parallel processing at hundreds of servers

Example: Web Search

- The average response time is 1.5 sec
- A study on Amazon and Google showed that an extra delay of 500 ms resulted in 1.2% loss of users and revenue



Image source : Google

This Course: Learning Contents and Approach

- Both theories and implementation skills
- **Theories**
 - DS design issues
 - Middleware technologies
 - Replication architectures
 - Fault tolerance technologies
 - Concurrency and Consistency control
 - Load balancing techniques
- **Implementation**
 - RMI (Remote Method Invocation) programming

Good Distributed System Design

- Principles for good decentralised algorithms
 - No machine has a complete state
 - Make a decision based on local information
 - A single failure does not bring down the system
 - No global clock
- Techniques
 - Asynchronous communication
 - Distribution of data and functions across multiple servers
 - Caching and replication

Akamai Technologies

- In 1990s, congestion control of internet was a big challenge
- In 1998, a content distribution network, Akamai Technologies, was founded by Thomson Leighton (Professor of Algorithms at MIT).
- Thomson Leighton's talk:
<https://www.youtube.com/watch?v=jMOwvQdTBbs&t=774s>
- Read the paper on the development of the modern internet:

The Akamai network: a platform for high-performance internet applications

<https://courses.cs.duke.edu/spring23/compsci512/papers/akamai-cdn.pdf>

Deciding the System Architecture

Question: Which system architecture is better, centralised or distributed?

Ans:

- It depends on your system properties.
- Suppose it requires only a small number of computers and all users stay close to a single location, then it is better to use a centralised system architecture. Otherwise, we use a distributed system.
- We need to evaluate the advantages and disadvantages of both architectures to judge their suitability.

Benefits of Distributed Systems

- **Rapid system development:** reuse existing components or services, e.g., e-Commerce (card verification), Google map
 - Study development technologies: middleware
- **Construct a low-cost and scalable system:** link up many low-cost machines to work together as a coherent system, e.g., Google search, Cloud computing
 - Study task distribution, synchronisation, concurrency and consistency control
- **Improve system reliability or availability:** based on replicated components, e.g., Online banking system
 - Study fault detection and tolerance, and error recovery

Question: Doing a Startup

Question: Suppose you want to do a startup and make it successful. It is better to consider a distributed system design. Is it true or false?

Ans: True.

- A distributed system architecture allows us to start the company with a smaller budget by renting servers or using services from existing providers.
- Also, you can scale the company up or down depending on its success or failure.

Transparency

Transparency: The phenomenon by which a distributed system attempts to hide:

- Its processes and resources are **physically distributed** across multiple computers, possibly separated by large distances, and **operated by different** companies.
- **Changes in components.**

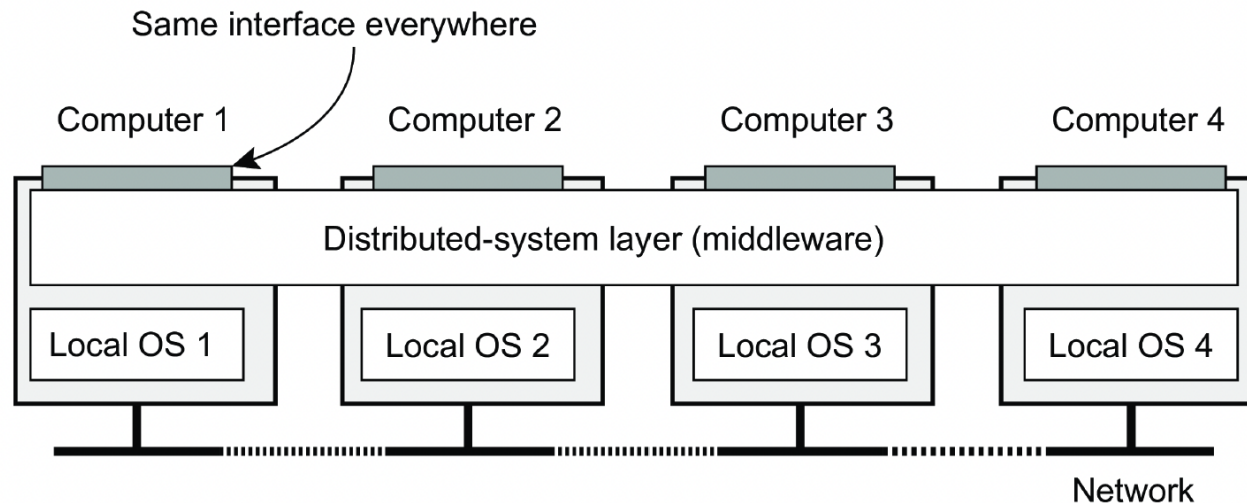
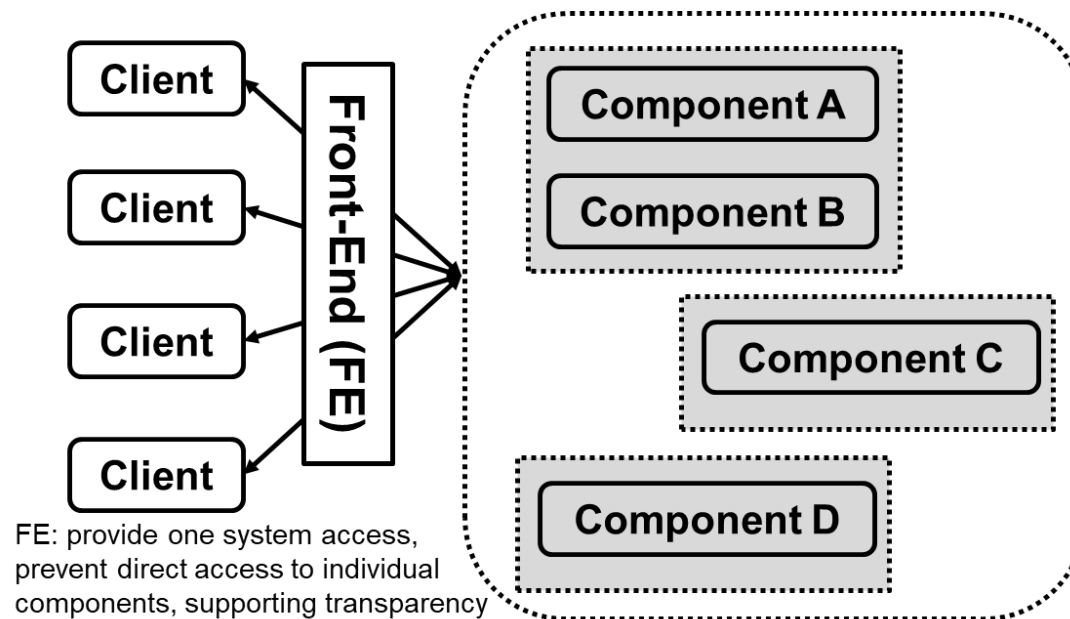


Image source: Steen & Tannenbaum textbook

Transparency

The front-end is used to achieve transparency; customers interact only with the front-end



NS Term 2 Lecture 1 / FL

Transparency Requirements – Good design of DS

Hide the separation of components or their organisation and changes from the user

Types of transparency	Description
Access	Hide differences in data representations and how a component is accessed, local and remote resources are accessed through identical operations, e.g., Middleware
Location	Hide where a component is located
Migration	Hide that a component or a resource may move to another location (content distribution network)
Relocation	Hide that a component or a resource may be moved to another location while in use (caching)
Replication	Hide that a component or a resource may have multiple copies (provide fast or local access)
Concurrency	Hide that a resource may be shared by several competitive users (handle shared resources)
Failure	Hide the failure and recovery of a resource or a component

Design Issues (1) – challenges to face

Network issues:

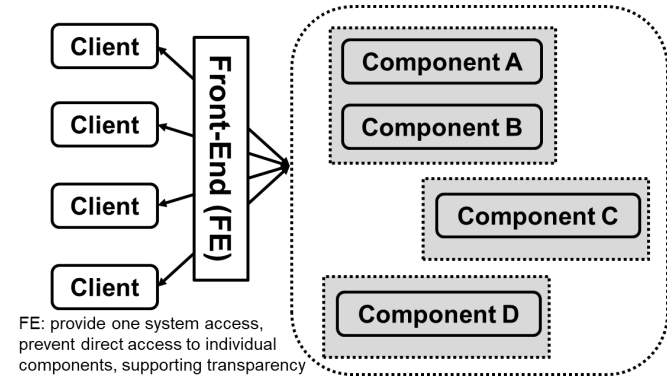
- Highly variable bandwidth
- Possibly large and variable latency

Distributed event issues:

- No native support for clock synchronisation among hosts
- Difficult to ensure data consistency

System failure issues:

- Unpredictable failures of components: may be due to the failure of a network component or communication delay incurred on the communication path
 - Software error in Microsoft systems:
<https://www.bbc.co.uk/news/articles/cp4wnrxqlewo>



NS Term 2 Lecture 1 / FL

Design Issues (2) – challenges to face

Security Issues:

- Centralised systems:
 - can rely on physical security
 - Users understand what trust to assign to the system
 - System administrators are responsible
- Distributed systems:
 - None of the above applies!
 - Hard to know what is being trusted or what can be trusted
 - Cyber attack on Microsoft Systems:
<https://www.bbc.co.uk/news/articles/c903e793w74o>

Reference

- **Chapter 1 of both books**



COMP2211

Distributed Systems

Topic 2: Middleware Technologies and RMI

Thiru Vasantam

This Topic

Aim

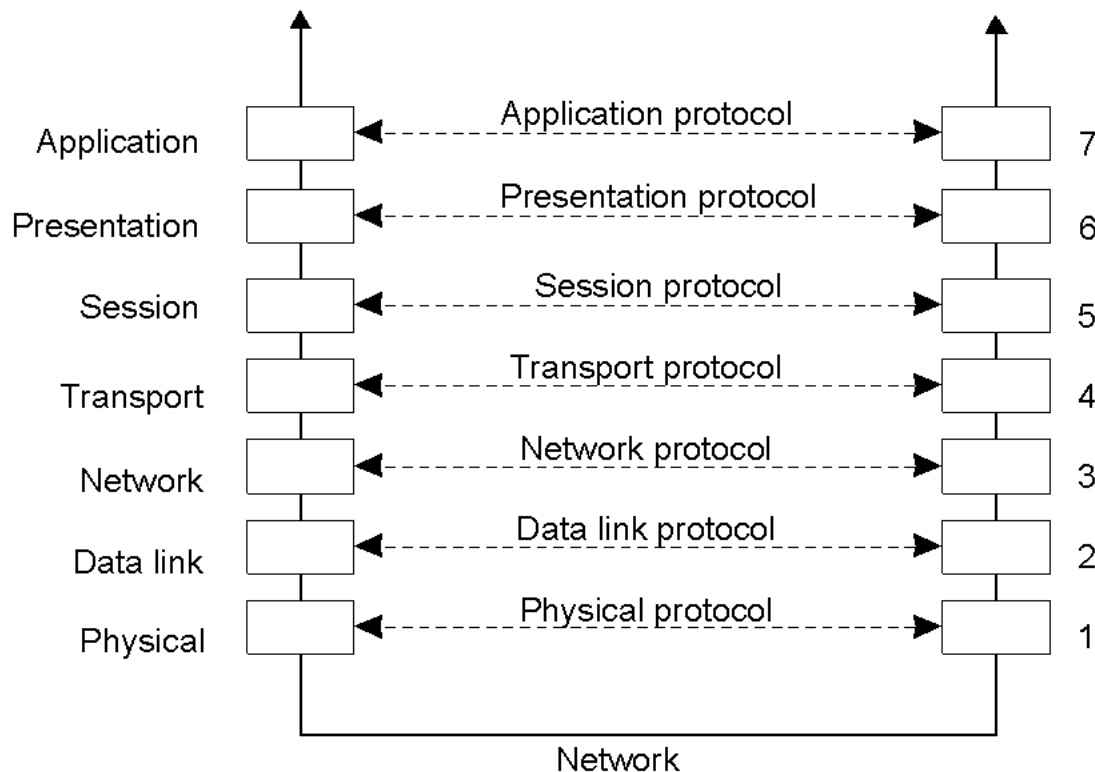
Study Middleware technologies

Understand how they support distributed systems development

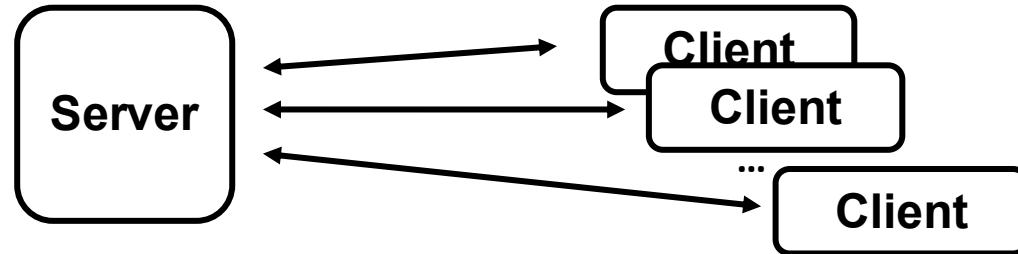
Study RMI technology

Communication Protocols

- Protocols are the agreements/rules on communication



Server and Client under Socket Programming



- Provide application functions
- Inputs to functions are incoming messages from clients
- Generate responses to clients
- Send messages to the server
- Receive responses from the server

References

- Chapter 4 (for RPC) and Chapter 2 (for RMI) of

*Distributed Systems, 4th Edition,
Maarten van Steen and Andrew
Tannenbaum.*

- Chapter 5 of

*Distributed systems : concepts and design /
George Coulouris ... [et al.], Addison-
Wesley (Pearson), c2012.*

